Planto.

Arsitektur & Teknologi Website E-commerce Tanaman

Dark mode UI | Glassmorphism | Next.js + PostgreSQL

Dokumen rancangan teknis berdasarkan desain UI landing page Planto

1. Analisis Section UI

Section	Komponen	Data
Navbar	Logo, menu, search, cart, hamburger	Static + cart count
Hero	Judul, CTA Explore, Live Demo, review card, featured product card	Featured product, 1 review
Our Trendy Plants	2 kartu horizontal (gambar bergantian kiri/kanan)	Produk trending
Our Top Selling	Grid 3x2 product card	Produk terlaris
Customer Review	3 kartu testimoni + rating bintang	Reviews
Our Best O2	Carousel dengan paginasi 01/04	Koleksi O2
Footer	Quick links, form newsletter, sosial	Subscribe API

2. Tech Stack

Frontend

- **Next.js 15 (App Router) + TypeScript**: SSR/SEO bagus untuk toko online
- **Tailwind CSS v4 + shadcn/ui**: styling dark theme dan komponen dasar
- **Framer Motion**: animasi hover kartu dan transisi hero
- **Embla Carousel**: section "Our Best O2"
- **Zustand**: state keranjang
- **TanStack Query**: fetching dan cache data
- **React Hook Form + Zod**: form newsletter dan checkout

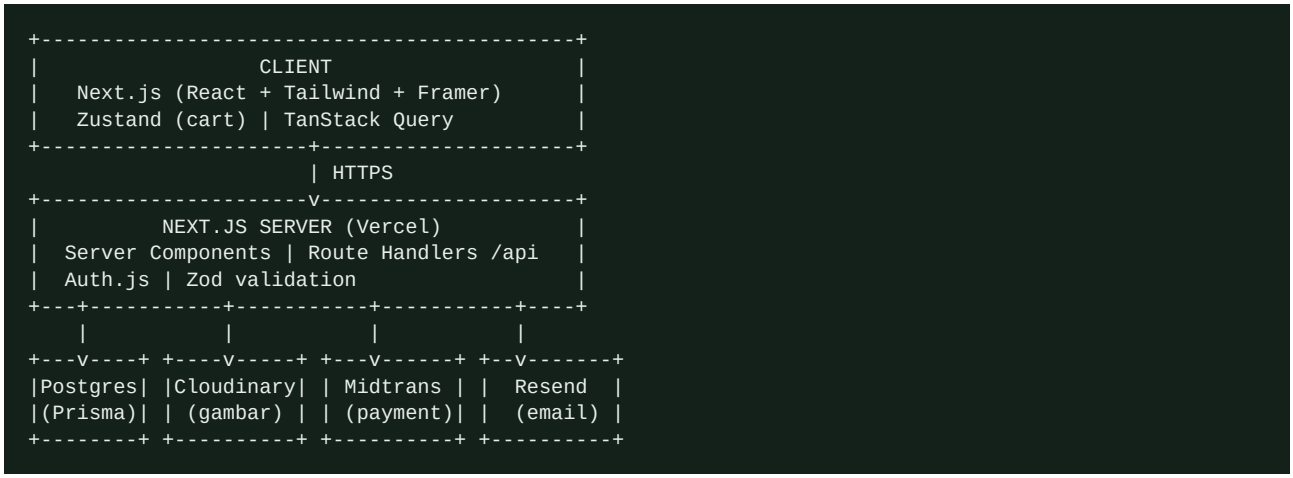
Backend

- **Next.js Route Handlers** (atau NestJS bila ingin dipisah)
- **PostgreSQL + Prisma ORM**
- **Auth.js (NextAuth)**: login Google/email
- **Midtrans atau Xendit**: pembayaran untuk pasar Indonesia
- **Resend**: email newsletter

Infrastruktur

- **Vercel**: hosting frontend
- **Neon / Supabase**: database
- **Cloudinary**: penyimpanan dan optimasi gambar tanaman (PNG transparan)
- **Sentry + Vercel Analytics**: monitoring

3. Diagram Arsitektur



4. Struktur Folder

```
planto/
|-- app/
|   |-- (shop)/
|   |   |-- page.tsx                # Landing page
|   |   |-- plants/page.tsx         # Katalog
|   |   |-- plants/[slug]/page.tsx
|   |   |-- cart/page.tsx
|   |   '-- checkout/page.tsx
|   |-- api/
|   |   |-- products/route.ts
|   |   |-- reviews/route.ts
|   |   |-- newsletter/route.ts
|   |   |-- cart/route.ts
|   |   '-- payment/webhook/route.ts
|   '-- layout.tsx
|-- components/
|   |-- layout/   (Navbar, Footer)
|   |-- sections/ (Hero, TrendyPlants, TopSelling,
|   |              CustomerReview, BestO2Carousel)
|   |-- ui/       (Button, GlassCard, RatingStars, SectionTitle)
|   '-- product/  (ProductCard, FeaturedCard)
|-- lib/          (prisma.ts, utils.ts, validators.ts)
|-- store/        cart-store.ts
|-- prisma/schema.prisma
'-- public/
```

5. Database (Prisma)

```
model Product {
  id          String   @id @default(cuid())
  slug        String   @unique
  name        String
  description String
  price       Int
  imageUrl    String
  isTrending  Boolean  @default(false)
  isTopSeller Boolean  @default(false)
  isO2Plant   Boolean  @default(false)
  stock       Int
  category    Category @relation(fields: [categoryId], references: [id])
  categoryId  String
  reviews     Review[]
}

model Category {
  id       String    @id @default(cuid())
  name     String
  products Product[]
}

model Review {
  id        String  @id @default(cuid())
  rating    Int
  comment   String
  user      User    @relation(fields: [userId], references: [id])
  userId    String
  product   Product @relation(fields: [productId], references: [id])
  productId String
}

model User {
  id      String   @id @default(cuid())
  name    String
  email   String   @unique
  avatar  String?
  reviews Review[]
  orders  Order[]
}

model Order {
  id     String      @id @default(cuid())
  userId String
  user   User        @relation(fields: [userId], references: [id])
  total  Int
  status String
  items  OrderItem[]
}

model OrderItem {
  id        String @id @default(cuid())
  orderId   String
  order     Order  @relation(fields: [orderId], references: [id])
  productId String
  qty       Int
  price     Int
}

model Subscriber {
  id    String @id @default(cuid())
  email String @unique
}
```

6. API Endpoints

Method	Endpoint	Fungsi
GET	/api/products?filter=trending|top|o2	Data tiap section
GET	/api/products/[slug]	Detail produk
GET	/api/reviews?limit=3	Testimoni
POST	/api/newsletter	Subscribe email
POST	/api/cart	Sinkron keranjang
POST	/api/checkout	Buat order + token Midtrans
POST	/api/payment/webhook	Update status pembayaran

7. Design Tokens (dari UI)

```
// tailwind.config
colors: {
  bg:      '#14201A',   // background gelap hijau
  surface: '#1F2D25',   // kartu
  border:  'rgba(255,255,255,0.12)',
  accent:  '#7BC043',   // hijau garis section title
  text:    '#E8EFE9',
  muted:   '#9DB0A3',
},
borderRadius: { card: '32px' },
fontFamily: { sans: ['Poppins', 'sans-serif'] },
```

Pola khas desain ini:

- **GlassCard**: `bg-white/5 backdrop-blur-md border border-white/10 rounded-[32px]`
- **SectionTitle**: judul dengan bracket hijau di sudut (pseudo-element)
- Gambar tanaman menyembul keluar dari kartu (`-mt-12` atau absolute positioning)

8. Catatan Penting

1. Harga di desain memakai **Rs.** (Rupee). Untuk Indonesia, ganti ke **IDR** dengan `Intl.NumberFormat('id-ID')`.
2. Teks "Breath Natureal", "Decs Ai Plat" dan "Our Best o2" masih typo; perbaiki menjadi "Breathe Naturally", "Deco Air Plant" dan "Our Best O2".
3. Semua kartu "Calathea plant" masih placeholder. Pastikan nama produk sesuai gambar, misalnya kaktus dan monstera.
4. Tambahkan state **responsive**: grid 3 kolom di desktop, 2 di tablet, 1 di mobile.